

Minkyong Kim

Ph.D. Candidate, Graduate School of Information, Yonsei University, Seoul, Korea
minky@yonsei.ac.kr | mfriendly.github.io | Google Scholar | github.com/mfriendly
ORCID 0009-0001-3530-0885

Research Interests

Machine learning for prediction and decision-making in dynamic environments: forecasting under distribution shift, counterfactual estimation of intervention effects, and selection of interventions under resource constraints. Methods draw on graph neural networks, time series models, and large language models, with applications in public health, energy, and economics.

Education

Yonsei University, Graduate School of Information <i>Integrated M.S./Ph.D. in Information Systems (expected February 2028)</i>	Seoul, Korea Sep. 2021 – Feb. 2028
• Advisor: Prof. Beakcheol Jang	
University of Pennsylvania, College of Arts and Sciences <i>B.A. in Economics</i>	Philadelphia, PA, USA May 2015

Publications

- [1] Beyond similarity: Coverage-aware prompt selection for time series forecasting with LLMs
D. Ji[†], **M. Kim**[†], D. Kim, Y. Lee, B. Kim*, B. Jang*
Findings of the Association for Computational Linguistics: EMNLP 2026
- [2] CTRL: Control-based time series forecasting with LLM-guided residual learning
M. Kim, D. Ji, Y. Lee, B. Kim, B. Jang
Findings of the Association for Computational Linguistics: ACL 2026
- [3] Forecasting epidemic spread with recurrent graph gate fusion transformers
M. Kim, J. H. Kim, B. Jang
IEEE Journal of Biomedical and Health Informatics, 29(2), 1546–1559, 2025
- [4] Accurate total consumer price index forecasting with data augmentation, multivariate features, and sentiment analysis: A case study in Korea
I. Seo[†], **M. Kim**[†], J. W. Kim, B. Jang
PLOS ONE, 20(5), e0321530, 2025
- [5] COVID-19 policies and payment delays: Panel data insights from Korea’s food industry
M. Kim, G. I. Kim, Y. H. Lee, B. Jang
International Conference on Artificial Intelligence in Information and Communication (ICAIIIC), 2025
- [6] An approach on improving the recommender system: Predicting movie genres based on plot summaries
G. I. Kim, J. H. Kim, **M. Kim**, T. Kwon, B. Jang
International Conference on Artificial Intelligence in Information and Communication (ICAIIIC), 2024
- [7] Time series decomposition and hybrid RNN-based spread prediction: Focused on the international energy futures market
M. Kim, S. Hwang, H. Y. Kim
The Korean Journal of Financial Management, 39(4), 89–122, 2022

[†] Equal contribution. * Corresponding authors.

Manuscripts Under Review and Preprints

[P1] Chance-constrained selection of sequential intervention strategies from counterfactual estimates

M. Kim, B. Jang

Under review. arXiv:2608.13209, 2026

[P2] Bilevel graph structure learning, revisited: Inner-channel origins of the reported gain

M. Kim, B. Jang

arXiv:2605.07577, 2026

[P3] AutoEpiGCL: Compartmentally structured graph contrastive learning for epidemic forecasting

M. Kim, B. Jang

Under review, 2026

Teaching

Incheon Academy of Science & Arts

Instructor, Data Analysis in Everyday Life

Incheon, Korea

Spring 2025

Yonsei University, Graduate School of Information

Teaching Assistant, Artificial Intelligence and Bigdata Research (GSI7657)

Seoul, Korea

Fall 2026

Yonsei University, Graduate School of Information

Teaching Assistant, Machine Learning (GSI7700)

Seoul, Korea

Spring 2026

Yonsei University, Graduate School of Information

Teaching Assistant, Bigdata Analytics and Applications (GSI7656)

Seoul, Korea

Fall 2025

Yonsei University, Graduate School of Information

Teaching Assistant, Database Strategy and Practice (GSI6001)

Seoul, Korea

Spring 2025

Academic Service

Conference reviewer: NeurIPS 2026 (Evaluations and Datasets Track), AAAI 2026

Journal reviewer: Travel Medicine and Infectious Disease

Professional Experience

Pebbles Investment

Assistant Manager, venture and alternative investment review

Seoul, Korea

Jan. 2019 – Dec. 2019

Hyundai Investment

Assistant Manager, Product Strategy Department; fund strategy simulation

Seoul, Korea

Jan. 2016 – Jun. 2017

Korea Institute of Finance

Research Intern, financial market policy research

Seoul, Korea

Jun. 2012 – Aug. 2012

Certifications and Skills

Certifications: Financial Risk Manager (FRM)

Programming: Python, PyTorch, R

Languages: Korean (native), English (fluent)